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Supplementary Material S2: UML-Lite Quick Start Guide

A Minimal UML Approach for Ecological Modelers

Introduction

This guide provides a streamlined introduction to UML for ecologists who want to improve model documentation without extensive software engineering training. The “UML-lite” approach focuses on two essential diagram types that provide maximum benefit with minimal learning investment.

Why UML-Lite?

Traditional Approach	UML-Lite Approach
Narrative text descriptions	Structured diagrams + brief text
Implicit assumptions	Explicit, visible assumptions
Difficult to compare models	Side-by-side visual comparison
No standard format	Standardized notation

Time investment: 2-4 hours to learn UML-lite basics

The Two Essential Diagrams

1. Class Diagram: "What exists in your model?"

The class diagram shows the **components** of your model and their **relationships**.

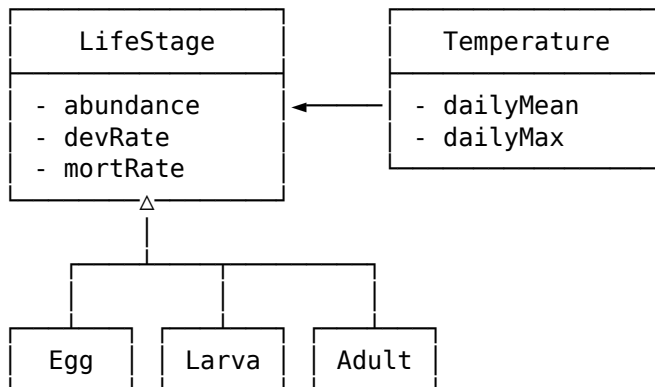
Basic Notation

ClassName	← Name of component
- attribute1 - attribute2	← Properties (variables) "-" means private
+ operation1() + operation2()	← Behaviors (functions) "+" means public

Relationship Symbols

Symbol	Meaning	Example
—>	Association (uses)	Larva uses Temperature
◇—	Aggregation (has)	BreedingSite has Eggs
◆—	Composition (contains)	Environment contains Temperature
△	Inheritance (is-a)	Larva is-a LifeStage
...>	Dependency (affects)	Larvicide affects Larva

Minimal Example: Simple Mosquito Model

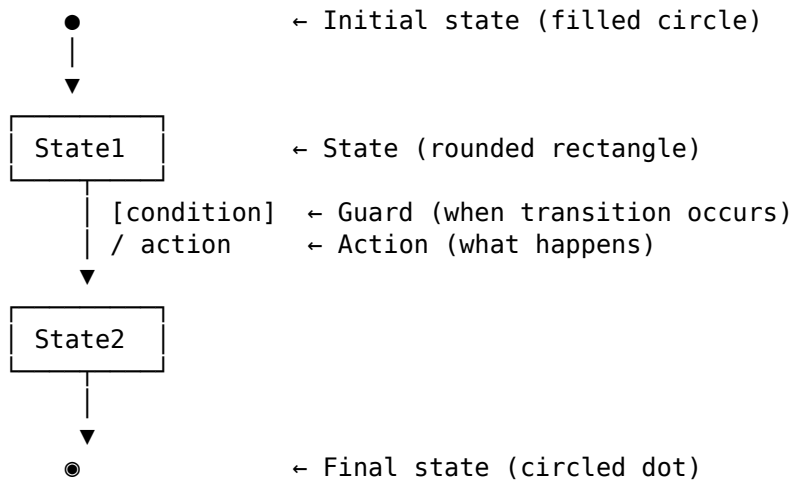


What this tells us: - Three life stages exist (Egg, Larva, Adult) - All stages share common properties (abundance, devRate, mortRate) - Temperature affects all life stages

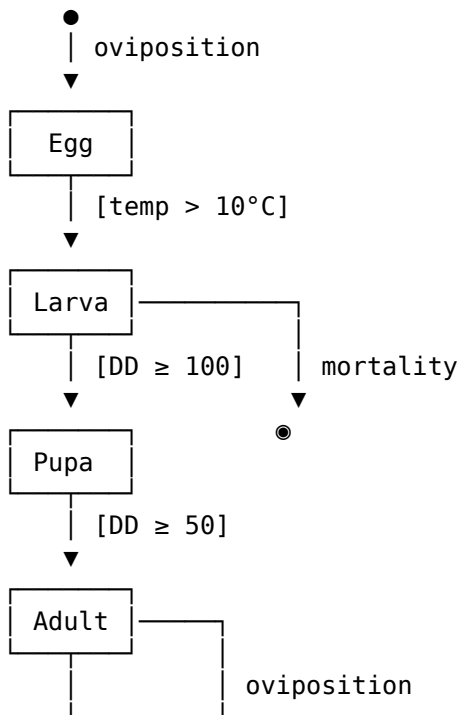
2. State Machine Diagram: "How does your model change over time?"

The state machine shows **states** (conditions) and **transitions** (changes between states).

Basic Notation



Minimal Example: Mosquito Life Cycle



What this tells us: - Life proceeds: Egg → Larva → Pupa → Adult → (lays eggs) - Hatching requires temperature > 10°C - Development tracked by degree-days (DD) - All stages can die (transition to final state)

PlantUML: Making Diagrams from Text

PlantUML lets you create diagrams by writing simple text. This enables: - Version control (track changes with Git) - Easy sharing and collaboration - Reproducible figures

Class Diagram in PlantUML

```
@startuml
class Egg {
    - abundance: float
    - diapause: boolean
    + hatch(): void
}

class Larva {
    - abundance: float
    - instar: int
    + moult(): void
}

class Temperature {
    - dailyMean: float
}

Egg --> Larva : develops into
Temperature --> Egg : affects
Temperature --> Larva : affects
@enduml
```

State Machine in PlantUML

```
@startuml
[*] --> Egg : oviposition

Egg --> Larva : [temp > 10]
Larva --> Pupa : [DD >= 100]
Pupa --> Adult : [DD >= 50]
Adult --> Egg : oviposition

Egg --> [*] : mortality
Larva --> [*] : mortality
Pupa --> [*] : mortality
Adult --> [*] : mortality
@enduml
```

Rendering Diagrams

Option 1: Online (no installation) - Go to: <https://www.plantuml.com/plantuml/> - Paste your code - Download the image

Option 2: VS Code extension - Install “PlantUML” extension - Preview diagrams as you type

Option 3: Command line

```
java -jar plantuml.jar mydiagram.puml
```

Checklist: Creating Your First UML-Lite Model

Class Diagram Checklist

- ☐ List all model components (state variables, drivers, interventions)
- ☐ For each component, list key attributes (properties)
- ☐ For each component, list key operations (behaviors)
- ☐ Draw inheritance relationships (is-a)
- ☐ Draw associations (affects, uses, contains)
- ☐ Add notes to explain non-obvious relationships

State Machine Checklist

- ☐ Identify all possible states in your model
- ☐ Mark the initial state(s)
- ☐ Mark terminal states (death, removal)
- ☐ Draw transitions between states
- ☐ Add guards: when does each transition occur?
- ☐ Add actions: what happens during transition?

Common Mistakes to Avoid

Mistake	Problem	Solution
Too much detail	Diagram becomes unreadable	Focus on essential structure
Missing relationships	Unclear how components interact	Review each pair of components
Vague guards	Unclear when transitions occur	Use specific conditions
No initial/final states	Unclear where system starts/ends	Always include entry/exit points
Inconsistent naming	Confusion between diagrams	Use same names everywhere

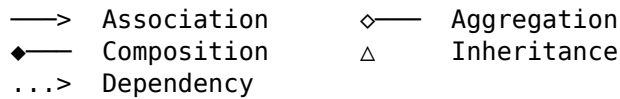
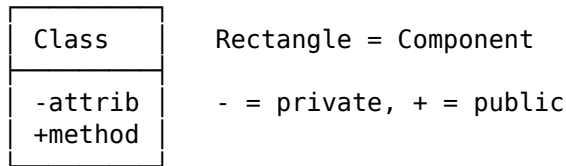
From UML-Lite to Full Framework

Once comfortable with UML-lite, consider adding:

Level	Diagrams	When to Use
Tier 1 (UML-Lite)	Class + State Machine	Basic documentation
Tier 2 (Standard)	+ Activity + Sequence	Formal specification, code generation
Tier 3 (Advanced)	+ OCL constraints, CI/CD	Regulatory models, team projects

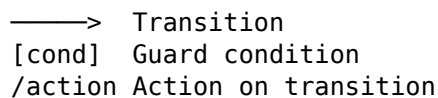
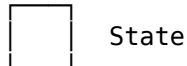
Quick Reference Card

Class Diagram Symbols



State Machine Symbols

- Initial state
- ⦿ Final state



PlantUML Quick Syntax

```
' Class diagram
class Name { }
ClassA --> ClassB : label

' State machine
[*] --> State1
State1 --> State2 : [guard] / action
State2 --> [*]
```

Resources

- **PlantUML website:** <https://plantuml.com/>
- **Online editor:** <https://www.plantuml.com/plantuml/>
- **UML specification:** <https://www.omg.org/spec/UML/>
- **This framework repository:** <https://github.com/sciom/aedes-uml-framework>

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